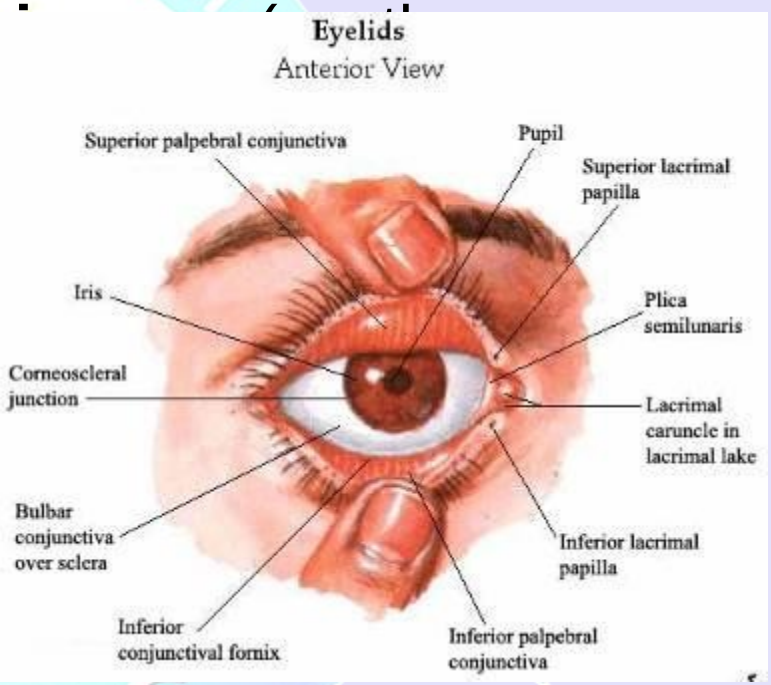
Eyelids, lacrimal apparatus and ciliary ganglion

Learning objectives

- At the end of the lecture the student should be able to know
- Eyelid and its parts
- Innervation and blood supply of eyelids.
- Parts of lacrimal apparatus.
- Diseases of lacrimal apparatus.
- Ciliary ganglion and its functions
- Disorders associated with ciliary ganglion.

Eye

- The **palpebral fissure** is the opening of the eye itself.
- The **lateral** and **medial con** angles) form the lateral and
- The medial commissure cor **(semilunar fold)** and the **la**



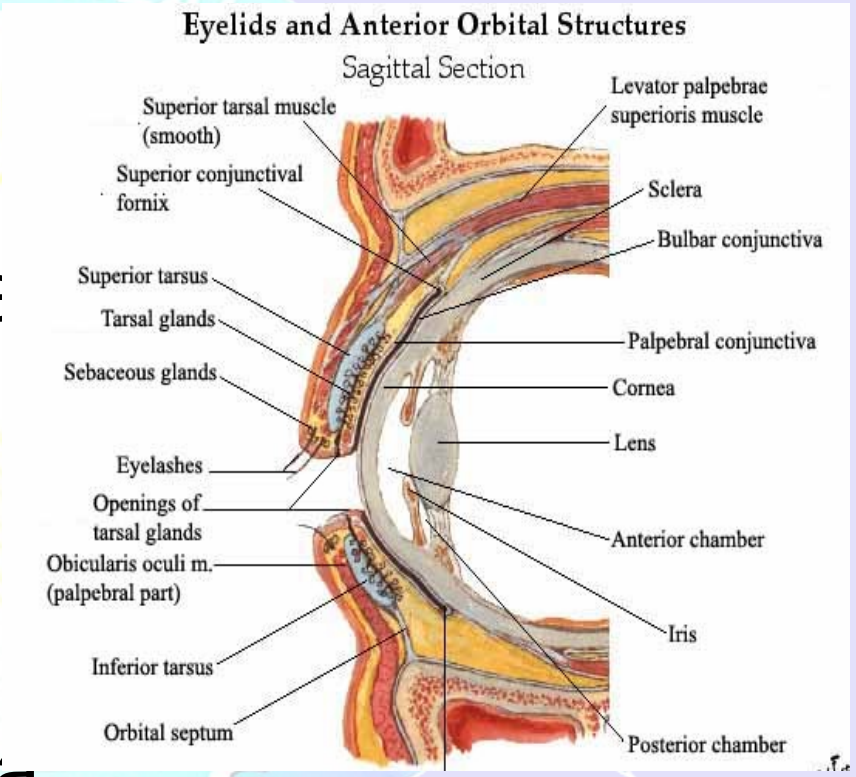
Eyelids

- **Eyelid** is a thin fold of skin that covers and protects an eye.
- Has thinnest skin of the whole bo
- The levator palpebrae superioris
  "open" the eye. This can be eithe
- Has a row of eyelashes which se
  the eye from dust and foreign de
- "Palpebral" (and "blepharo") mea



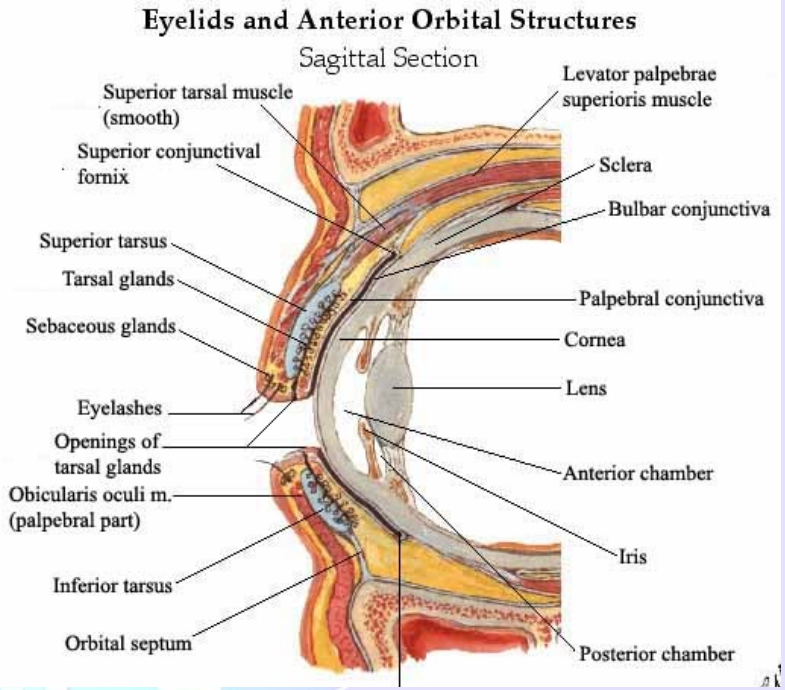
Layers of eyelids

- Eyelid is made up of
- Skin
- Subcutaneous tissue
- Orbicularis oculi
- Orbital septum
- Tarsal plates
- Palpebral conjunctiva



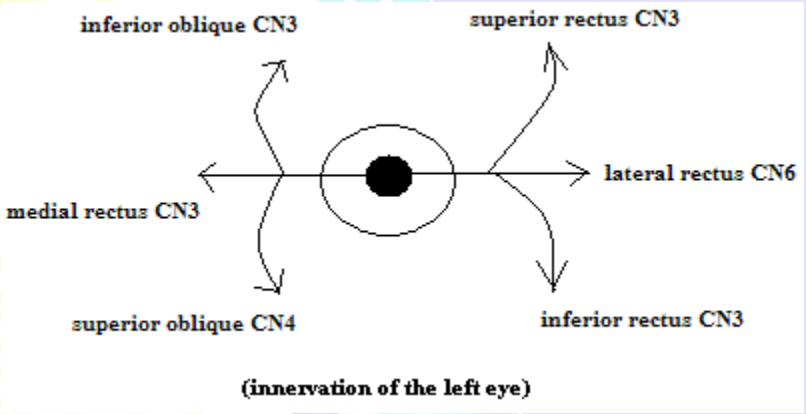
Eyelids

- The skin is similar to areas elsewhere, but has more pigment cells
- **Superior** and **inferior tarsal plat** give support and form to the eyeli secretions to prevent tears from le dry.
- **Orbicularis oculi muscle** contai and acts to close the eyelids.



Innervation of eyelids

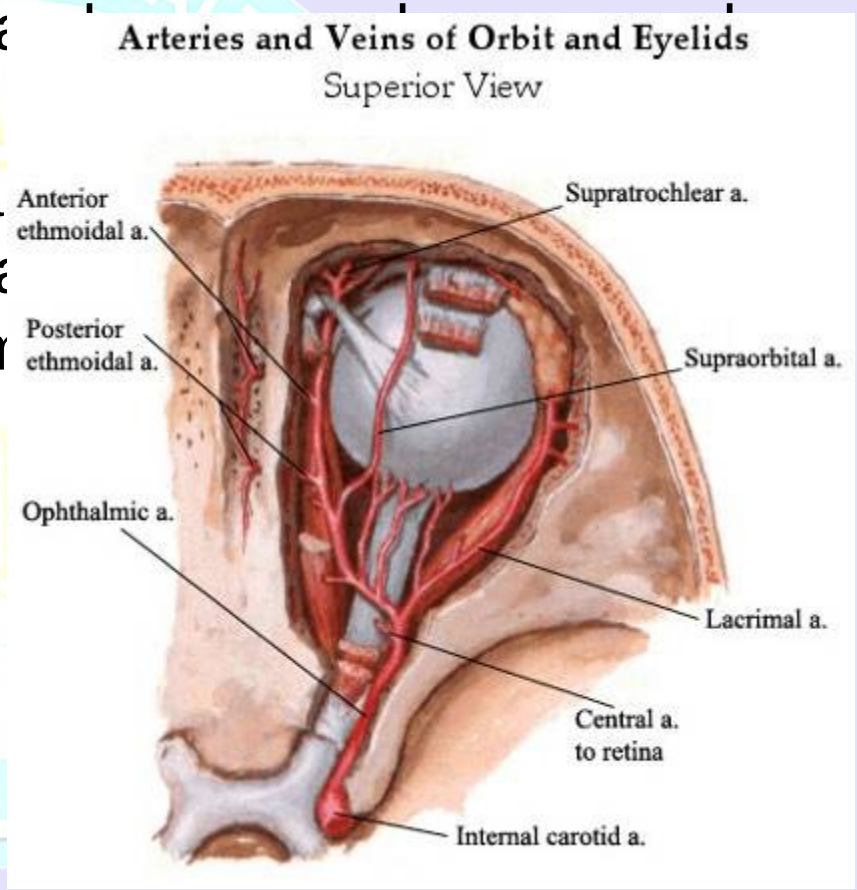
- Sensory nerve supply to the upper eyelids is from the infratrochlear, supratrochlear, supraorbital and the lacrimal nerves from the ophthalmic branch (V1) of the trigeminal nerve (CN V).
- The skin of the lower eyelid is supplied by branches of the infratrochlear at the medial angle, the rest is supplied by branches of the infraorbital nerve of the maxillary branch (V2) of the trigeminal nerve.



(innervation of the left eye)

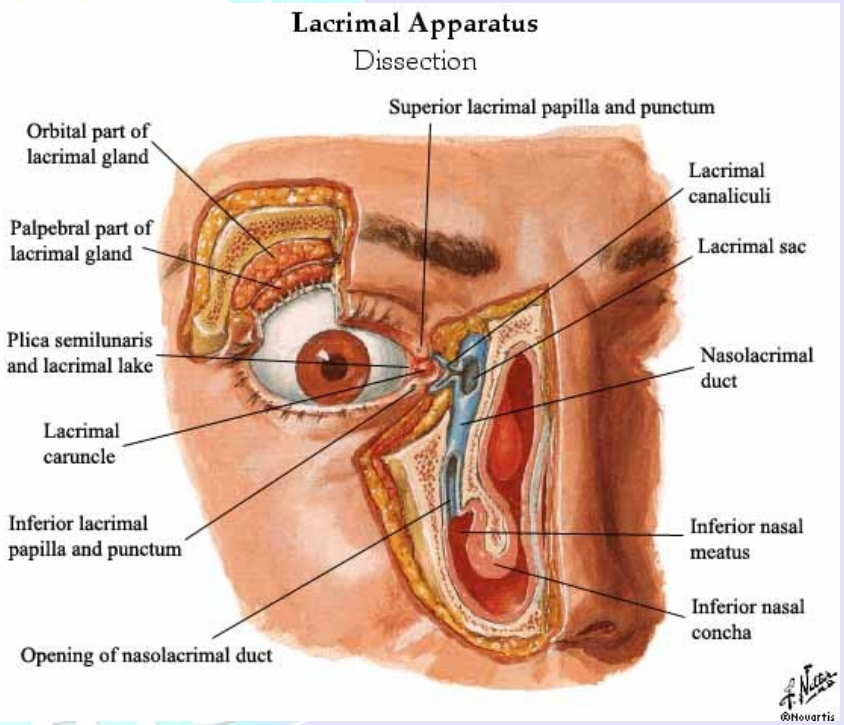
Blood supply of eyelids

- Supplied with blood by two a
  lower lid.
- The arches are formed by a
  palpebral arteries and media
  branching off from the lacrim
  artery, respectively.



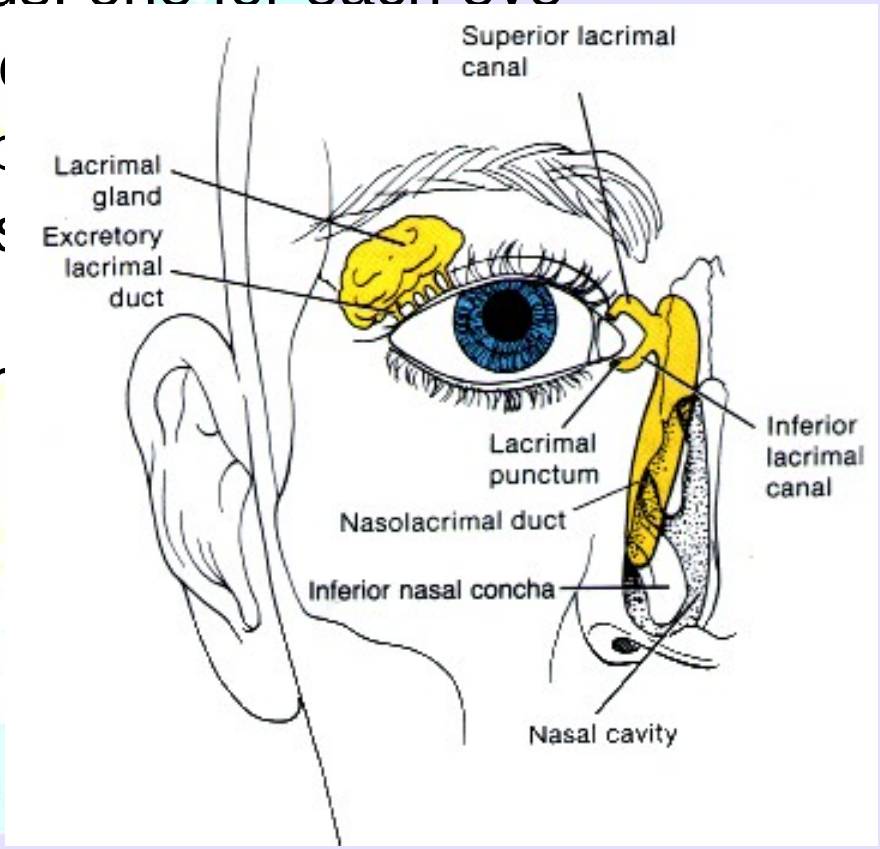
Lacrimal apparatus

- A system containing the orbital structures for tear production and drainage.
- Consists of:
  - Lacrimal gland
  - Lacrimal canaliculi
  - Lacrimal sac
  - Nasolacrimal duct



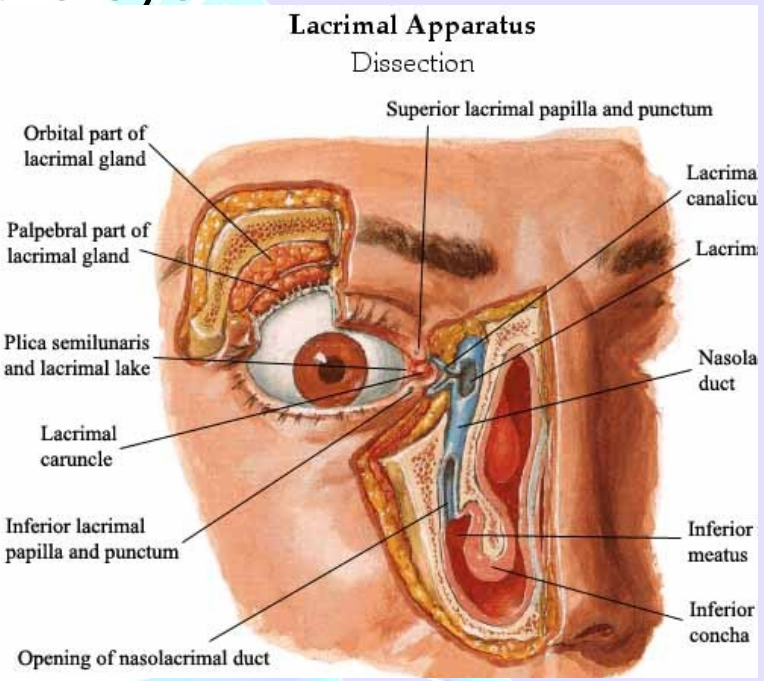
Lacrimal gland

- Paired almond-shaped glands, one for each eye
- Secretes the aqueous layer
- Situated in the upper, outer p
- It is lodged in the lacrimal fos
  the frontal bone.
- Has palpebral portion and or



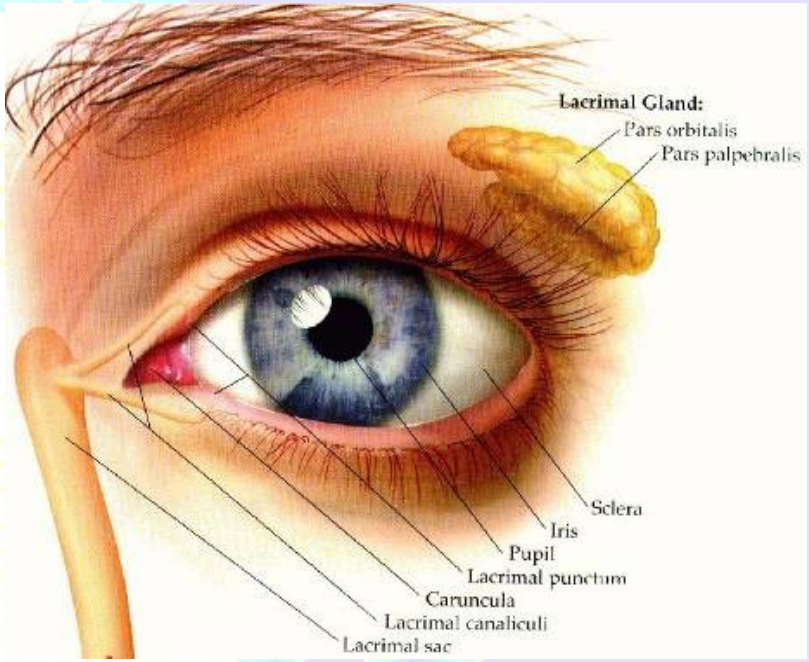
Lacrimal gland

- Palpebral portion lies close to the eye.
- Orbital portion contains fine in
  to form 3 - 5 main excretory du
  the palpebral portion before th
  on the surface of the eye.



Lacrimal puncta and canaliculi

- There are two lacrimal puncta in the medial (inside) portion of each eye.
- Collect tears and conveys to lacrimal sac through lacrimal canaliculi.
- Lacrimal canaliculi are small channels in each eyelid that commence at lacrimal puncta.
- *superior duct*, the smaller and shorter of the two, at first ascends, and then bends at an acute angle, and passes medialward and downward to the lacrimal sac.
- *inferior duct* at first descends, and then runs almost horizontally to the lacrimal sac.
- At the angles they are dilated into ampullæ



Lacrimal sac

- **Lacrimal sac** is the upper dilated end of the nasolacrimal duct
- Lodged in a deep groove formed by the lacrimal bone and frontal process of the maxilla.
- It connects the lacrimal canaliculi, which drain tears from the eye's surface, and the nasolacrimal duct, which conveys this fluid into the nasal cavity.
- oval in form
- measures from 12 to 15 mm. in length
- upper end is closed and rounded
- lower end is continued into the nasolacrimal duct.

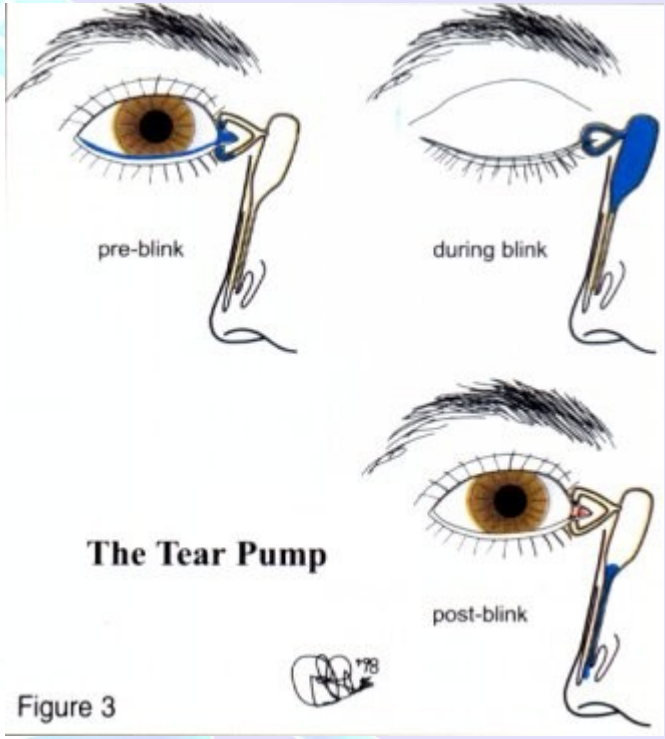
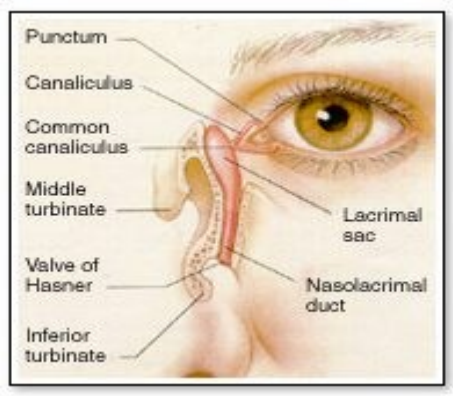
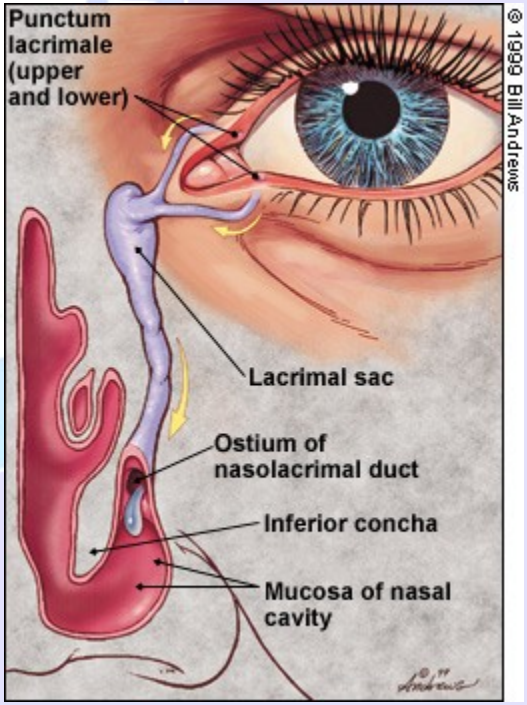


Figure 3

Nasolacrimal duct

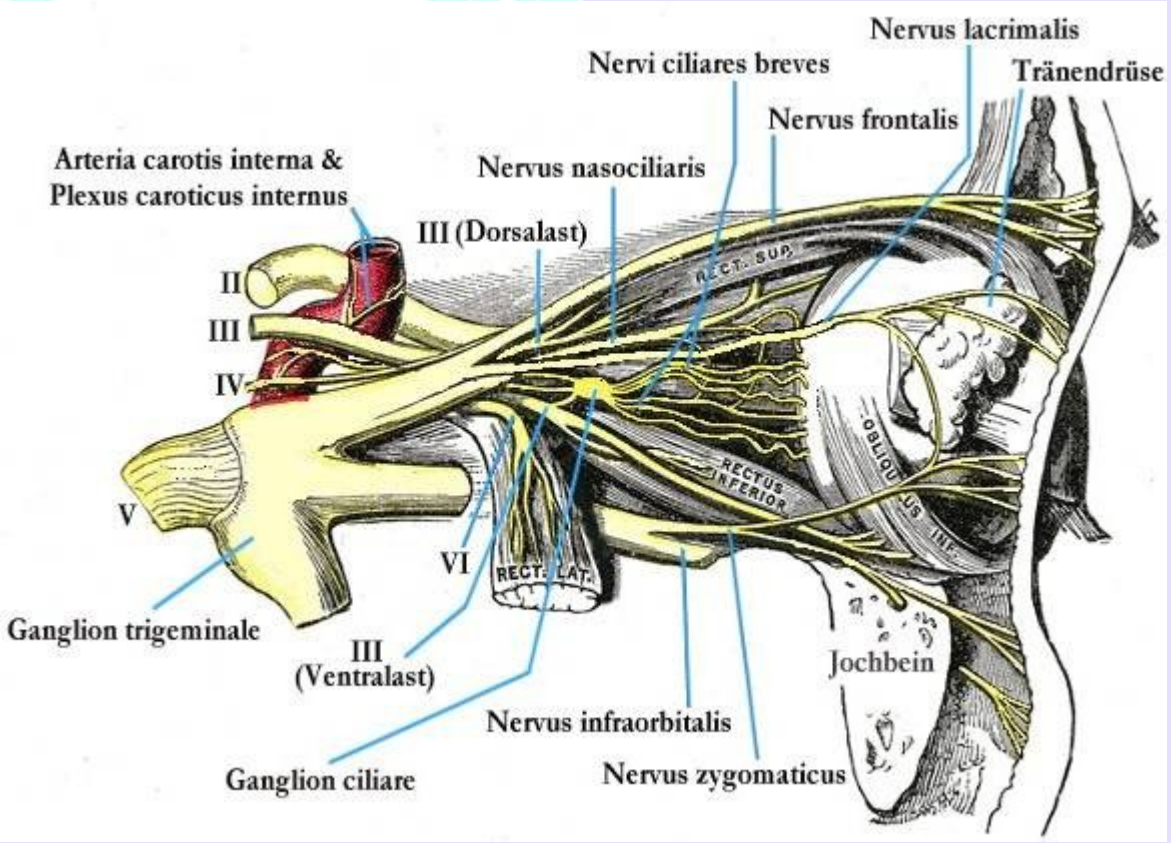
- Carries tears from the lacrimal sac into the nasal cavity. Excess tears flow through nasolacrimal duct which drains into the inferior nasal meatus.
- Duct is lined by stratified columnar epithelium containing mucus-secreting goblet cells, and is surrounded by connective tissue.
- Obstruction of the nasolacrimal duct leads to the excess overflow of tears called *epiphora*





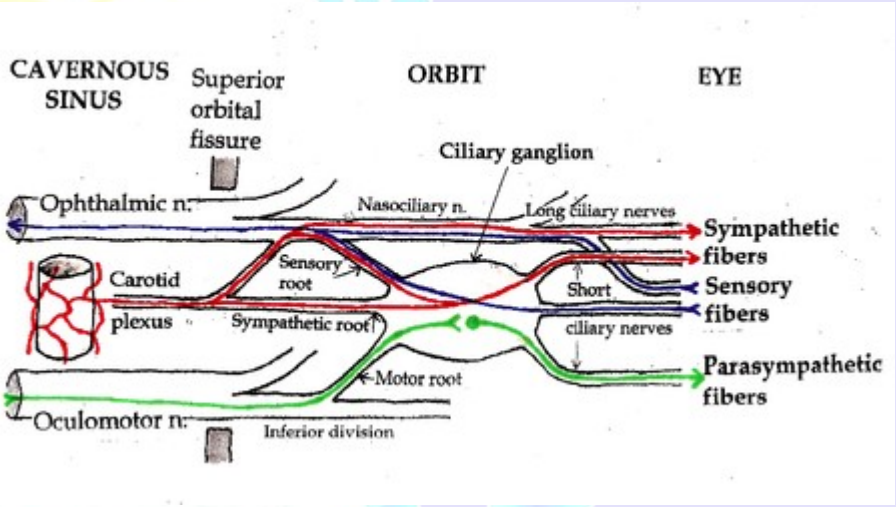
Ciliary ganglion

- C**iliary ganglion** is a **parasympathetic** ganglion located in the posterior orbit.
- It measures 1–2 millimeters in diameter and contains approximately 2,500 neurons.



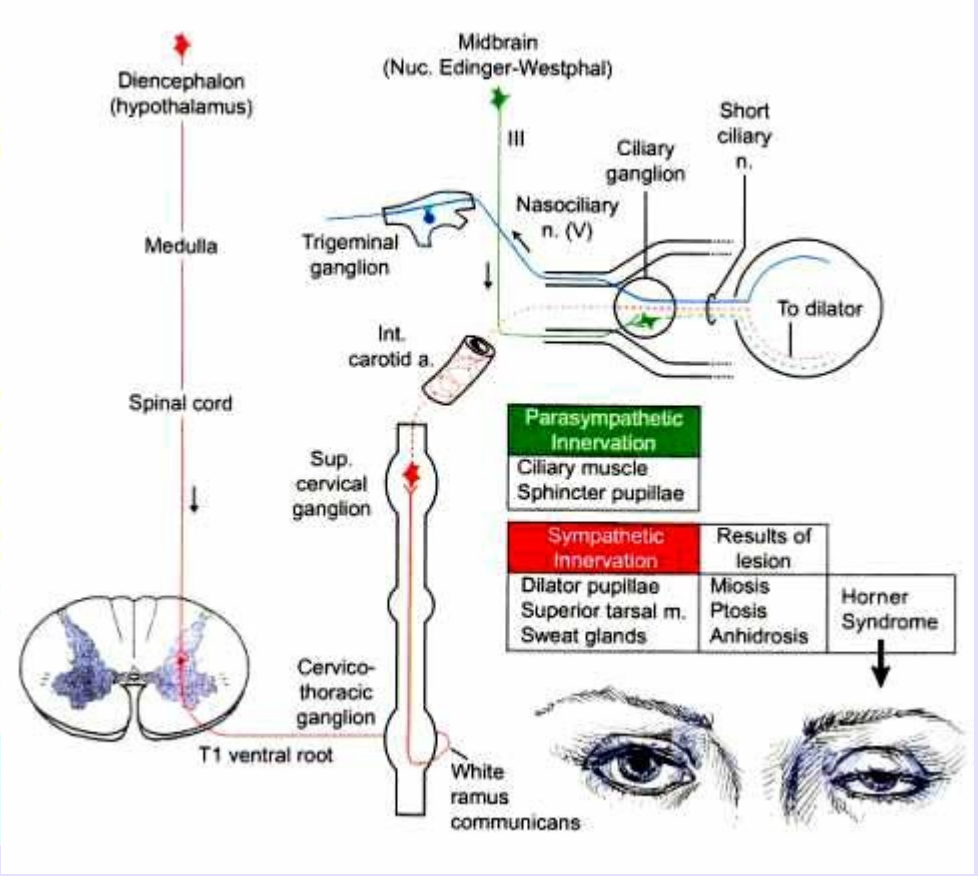
Fibres in ciliary ganglion

- Three types of nerve fibers run through the ciliary ganglion:
  - *parasympathetic fibers*
  - *sympathetic fibers*
  - *sensory fibers*.
- Only parasympathetic fiber
ganglion
- The other two types of nerv



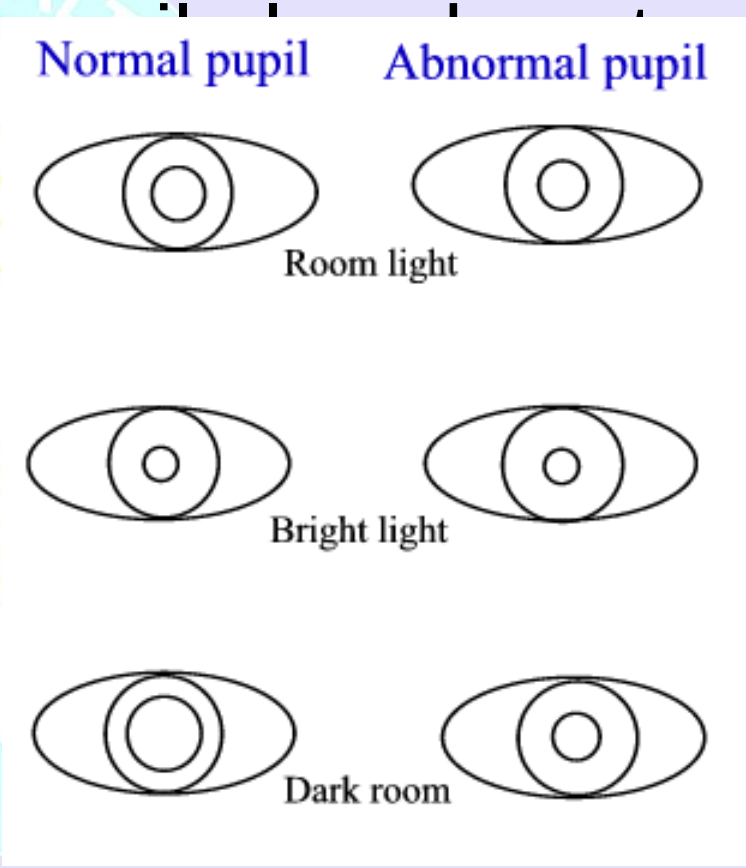
Function of ciliary ganglion

- Ciliary nerve innervates two muscles
- Sphincter pupillae
  - constricts the pupil, a movement known as Miosis.
- Ciliaris muscle
  - Releasing tension on the , making the lens more convex, also known as accommodation
- Both of these muscles are *involuntary* – they are controlled by the autonomic nervous system.



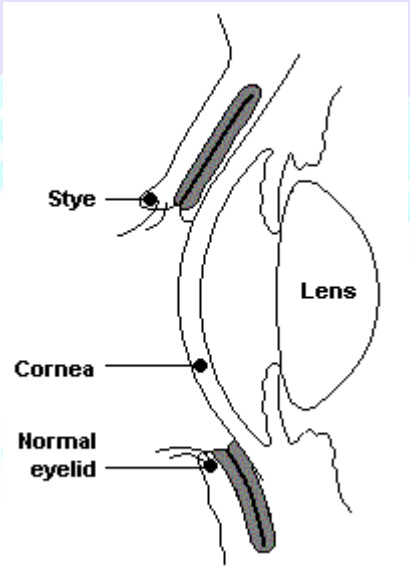
Adie syndrome

- **Adie syndrome** is tonic deep tendon reflexes.

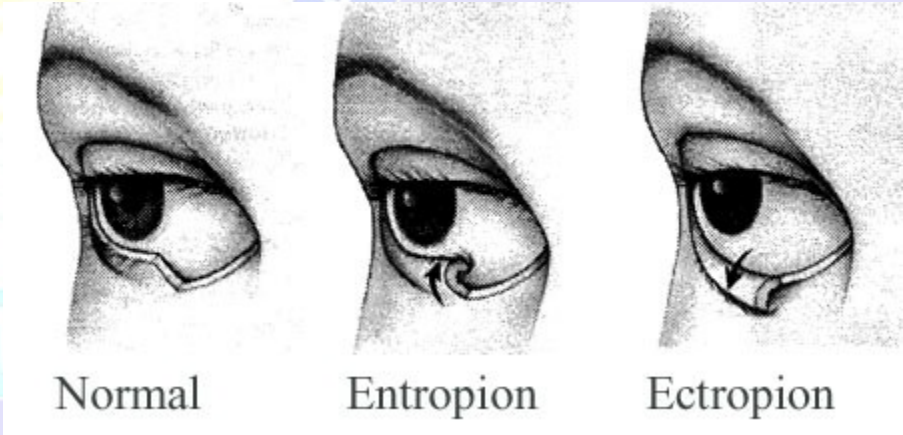


Eyelid disorders

- Styes
  - infections of the sebaceous glands of Zein usually caused by Staphylococcus Aureus bacteria.
- Entropion
  - It mostly affects the lower lid and it is characterized by the turning inward of the lid, toward globe.





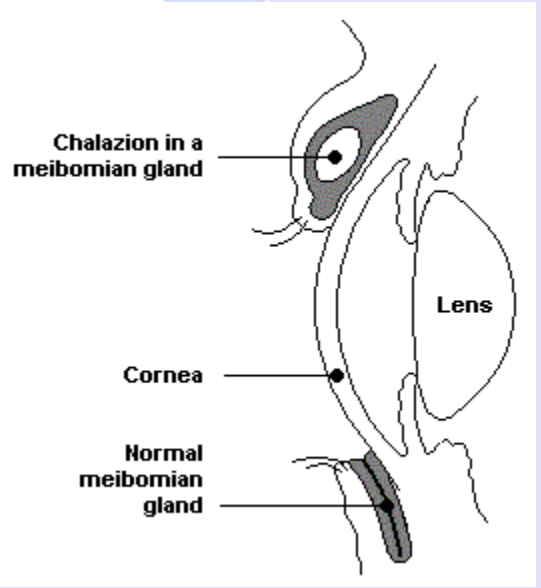


Eyelid disorders

- Chalazion
  - caused by the obstruct
    and can occur in both
    eyelids.

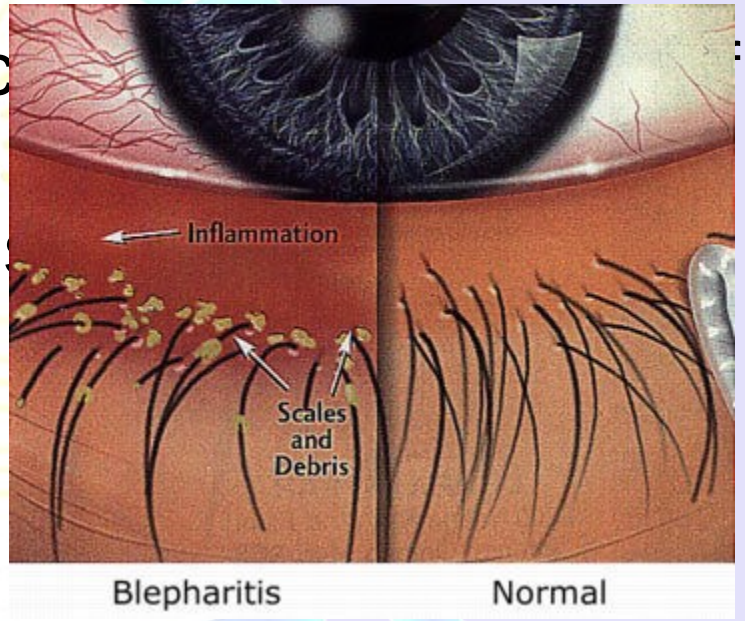


Chalazion



Eyelid disorders

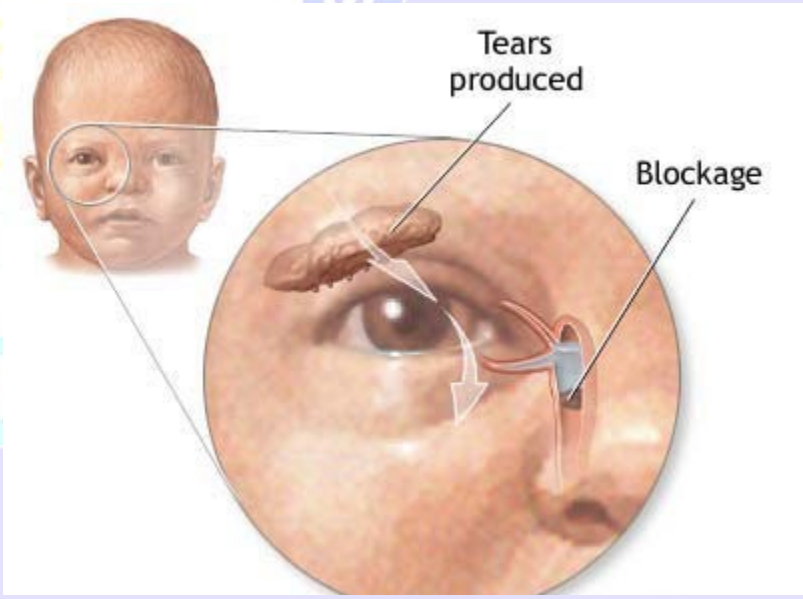
- Blepharitis
  - common condition that c
    the eyelids and which is
    manage because it tend



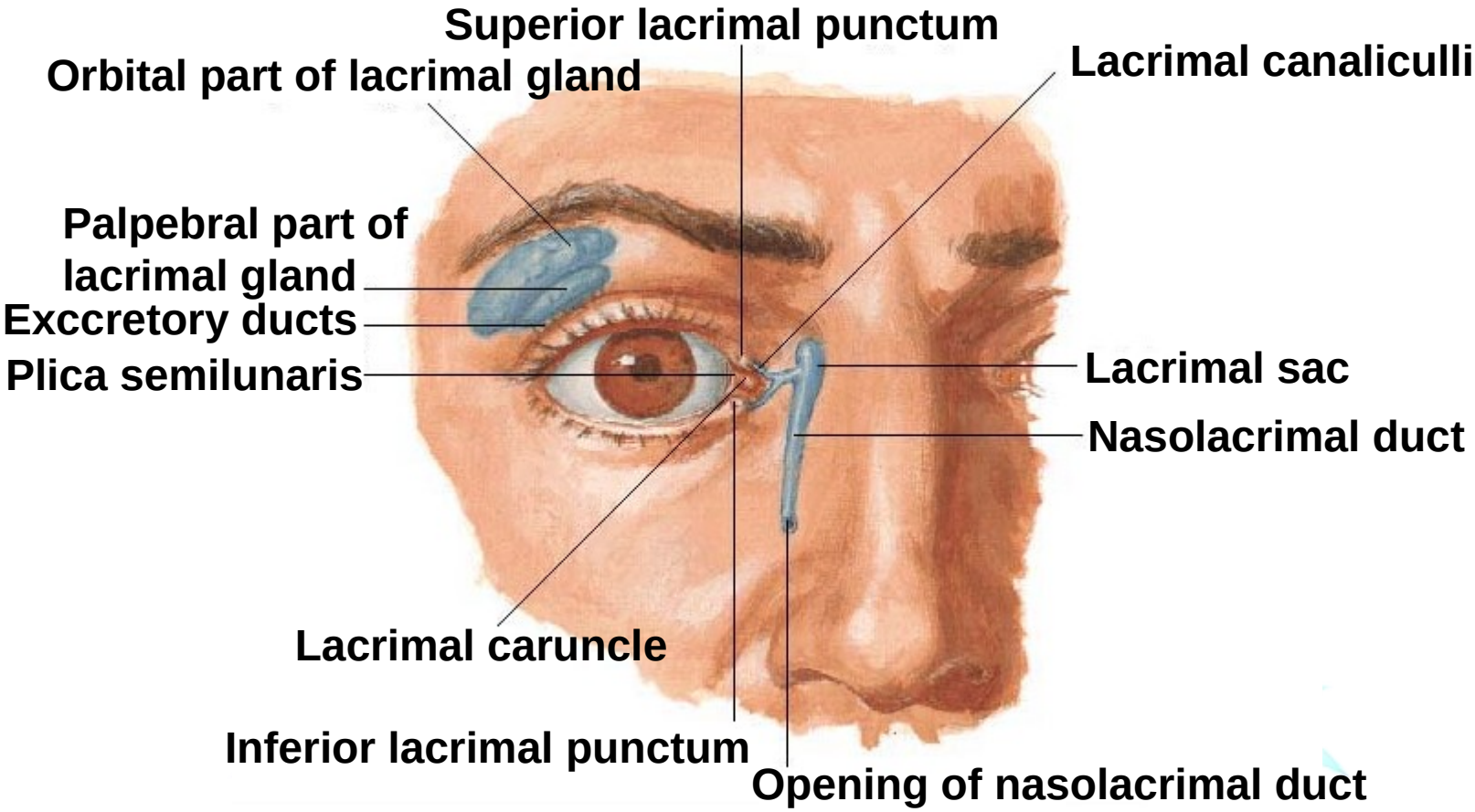
Lacrimal diseases

- **Dacryoadenitis** is infla
  lacrimal glands





Identify???


Superior lacrimal punctum
Orbital part of lacrimal gland
Lacrimal canaliculli
Palpebral part of lacrimal gland
Exccretory ducts
Plica semilunaris
Lacrimal sac
Nasolacrimal duct
Lacrimal caruncle
Inferior lacrimal punctum
Opening of nasolacrimal duct

Thanks